Name:

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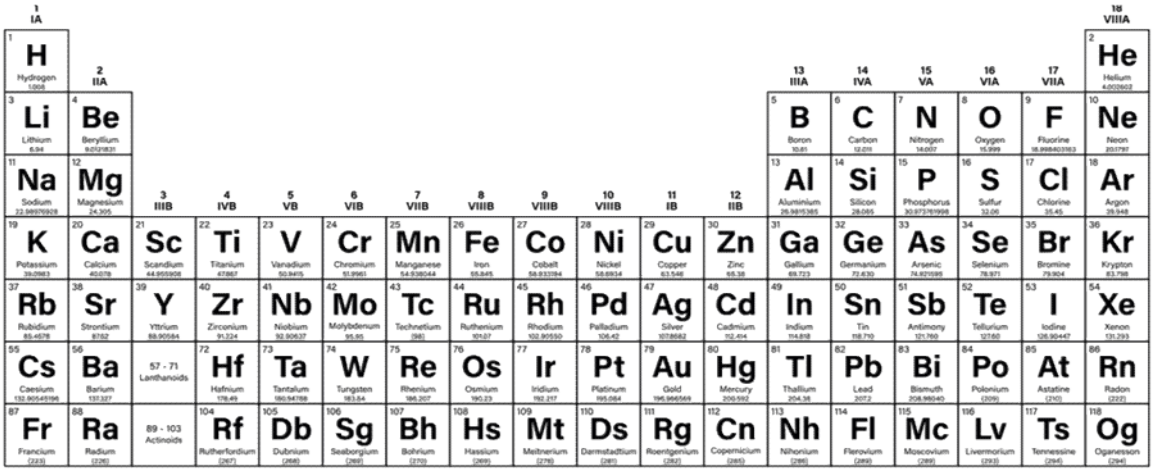
Instructions for CM2021 Mid-Term Test

1. This test paper contains Twenty (20) multiple-choice questions

2. Answer ALL questions (1 mark each)

3. Time allowed: 1 hour

4. Do not turn over the test paper until told to do so



1) Give the chemical name of $[Fe(CN)_2(CH_3NH_2)_4]$.

a. dichlorotetramethylamineiron(II)
b. dicyanotetrakis(dimethylamine)iron(II)
c. dicholortetramethylamineiron(IV)
d. dicyanotetrakis(methylamine)iron(II)

2) What is the coordination number of the metal center in $KMnO_4$?

a. 4
b. 5
c. 8
d. None of the above

3) What is the oxidation state of the nickel metal centers in $[Ni_2Cl_2(PPh_3)_2]$?

a. 0
b. +1
c. +2
d. +4

4) Which of the following compound shows only linkage isomerism?

a. $[Co(NH_3)_5(SO_4)](NO_2)$
b. $[Co(NCS)(NH_3)_3]SCN$
c. $[Co(en)_2(SCN)_2]$
d. $[Co(NO_2)Br(NH_3)_4]Cl$

5) How many isomers are possible for the complex $[Pt(en)(NO_3)(SCN)]$?

a. 2
b. 8
c. 12
d. None of the above

6) Which of the following is the correct name for $[Cu(H_2O)_6][Pt(CN)_6]$:

a. Hexaaquacopper(II) hexacyanoplatinate(II)
b. Hexaaquacopper(II) hexacyanoplatinate(IV)
c. Hexaaquacopper(I) hexacyanoplatinate(II)
d. Hexaaquacopper(II) hexacyanoplatinum(IV)

7) How many distinct isomers are possible for a metal complex containing three glycinato ligands?

a. 1
b. 2
c. 4
d. None of the above

8) Which of the following gives the highest number of isomers?

a. $[Co(NH_3)_4Cl_2]$
b. $[Ni(en)(NH_3)_4]^{2+}$
c. $[Ni(C_2O_4)(en)_2]$
d. $[Cr(SCN)_2(NH_3)_4]^{4+}$

9) Which of the following tetrahedral metal complex shows optical isomerism? M denotes a metal center. A, B, C, and D indicate different types of donor atom. A hypen "-" indicates bond between donor atoms (i.e., multidentate ligand).

a. $[M(A\text{-}A)_2]$
b. [M(A-A)(B-B)]
c. [M(A-A)(B-C)]
d. [M(A-B)(C-D)]

10) What is the name for $Na_2[Fe(C_2O_4)_2(H_2O)_2]$?

a. Sodium diaquabis(oxalato)iron(III)
b. Sodium diaquabis(oxalato)ferrate(II)
c. Disodium diaquabis(oxalato)ferrate(II)
d. Disodium diaquabis(oxalato)iron(II)

11) In ligand field theory, which of the following metal atomic orbitals are used for sigma bonding with ligands in an octahedral complex ?

a. No use of metal orbitals
b. all d orbitals
c. d_{xy}, d_{yz}, d_{xz} orbitals
d. d_{x2-y2}, d_{z2} orbitals

12) Which of the following statement indicates a stable metal complex?

a. Low formation constant
b. High dissociation constant
c. Weak metal-ligand binding affinity
d. None of the above

13) In the metal complex $[Zn(F)_6]^{4-}$, what type of bonding interaction(s) is/are present in the Zn-F bond?

a. Ligand-to-metal π character only
b. σ character only
c. Both ligand-to-metal and metal-to-ligand π characters
d. Both (a) and (b) are present

14) Which of the orbital will be most destablized when we convert a square planar metal complex into a square pyramidal metal complex by adding one additional ligand to the metal complex?

a. d_{xz}
b. d_{yz}
c. d_z^2
d. d_{xy}

15) What is the electronic arrangement of metal atom/ion in octahedral metal complex with d^4 configuration? Assume pairing energy < Δ_o

a. $t_{2g}^4 e_g^0$
b. $e_g^4 t_{2g}^0$
c. $t_{2g}^3 e_g^1$
d. $e_g^2 t_{2g}^2$

16) Which of the following octahedral metal complex exhibits the strongest Jahn-Teller effect?

a. High-spin d^4
b. Low-spin d^4
c. High-spin d^5
d. Low-spin d^5

17) Which of the following on phosphine ligand is true?

a. σ-donor, π-acceptor
b. σ-donor, π-donor
c. σ-acceptor, π-acceptor
d. None of the above

18) Identify the first-row transition metal M that satisfies the requirements given (more than one answer may be possible).
$K_2[MF_4]$ having half-filled doubly degenerate e orbitals.

a. Ti
b. Fe
c. Co
d. Both (b) and (c) are correct

19) What is the CFSE of a d^8 octahedral metal complex?

a. $1.2\Delta_o$

b. $1.2\Delta_o$ - P

c. $1.2\Delta_o$ - 2P

d. $1.2\Delta_o$ - 3P

20) The aqua ligands of $[Cu(H_2O)_6]^{2+}$ can be replaced by ammine ligands. Which of the following is expected to be the major product?

a. $[Cu(NH_3)_2(H_2O)_4]^{2+}$

b. $[Cu(NH_3)_4(H_2O)_2]^{2+}$

c. $[Cu(NH_3)_6]^{2+}$

d. None of the above

End of test paper.